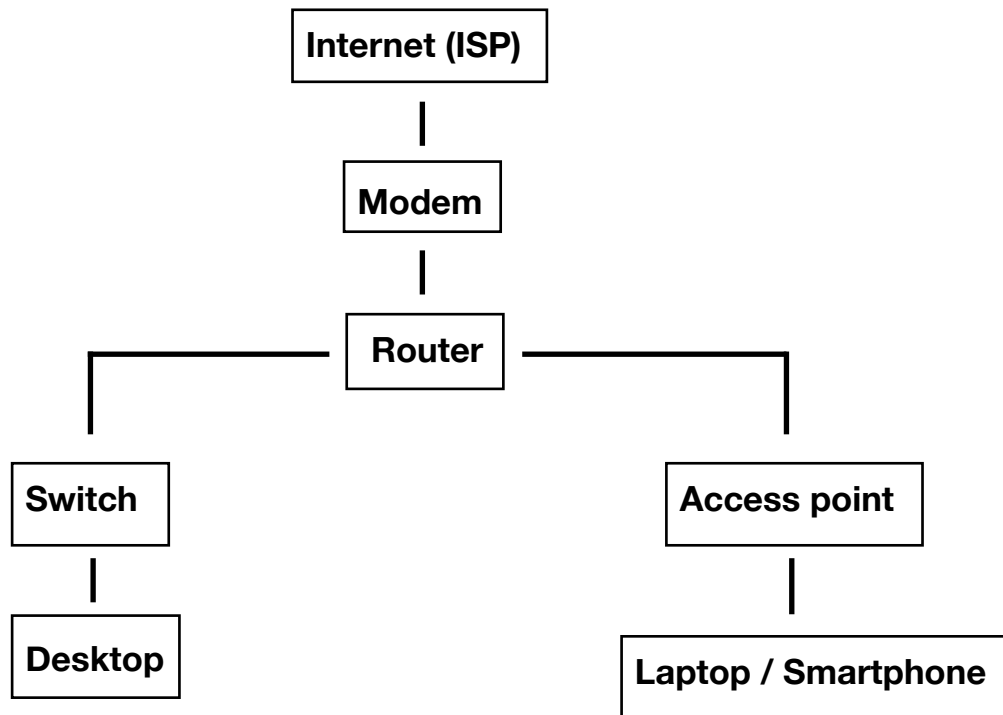


Module 3: A+ Understanding and Maintenance of Networks

DAY 1:-

Task 1:



Task 2:

Networking Hardware Devices

Modem

Converts the digital signal from your computer into a form that can travel over your ISP's line (cable, DSL, fiber) and back again.

Example: The box your internet provider installs to bring internet into your house.

Router

Directs data traffic between devices on your local network and the internet; assigns each device a local IP address.

Example: At home, to share one internet connection across multiple devices.

Switch

Connects multiple wired devices within the same local network, forwarding data only to the specific device it's meant for.

Example: In an office, connecting several desktop computers and a printer to the same network via cables.

Access Point

Extends a wired network by broadcasting a Wi-Fi signal, letting wireless devices join the network.

Example: In a large house or warehouse, placed in a far room to extend Wi-Fi coverage beyond the router's range.

Network Cable

Physically carries data signals between devices (computers, switches, routers).

Example: An Ethernet cable running from your desktop PC to a wall jack or router for a stable, fast connection.

Task 3:

1. Ethernet Cable



Commonly used to connect computers, routers, and switches within a home or office LAN.

2. Coaxial Cable



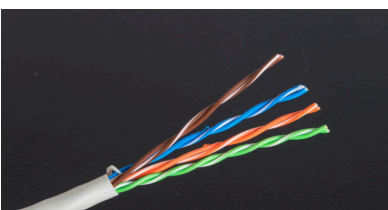
Commonly used to connect cable TV boxes and cable internet modems to the wall outlet.

3. Fiber Optic Cable



Commonly used for high-speed, long-distance connections such as ISP backbone links , data centres etc.

4. Twisted pair Cable



Commonly used in Home/Office Ethernet Networks, LANs, Security Cameras etc.

Task 4:

	Router	Switch
Main Use	Connects your local network to the internet and routes traffic between networks	Connects multiple wired devices within the same local network
Location	Where the internet enters the home/office (near the modem)	Wherever many wired devices need connecting, e.g. an office floor
Data handling	Uses IP addresses to route data between different networks	Uses MAC addresses to forward data to the correct device on the LAN
Network Layer	Operates at Layer 3 (Network layer)	Operates at Layer 2 (Data Link layer)